Bistability of Atmospheric Circulation on TRAPPIST-1e

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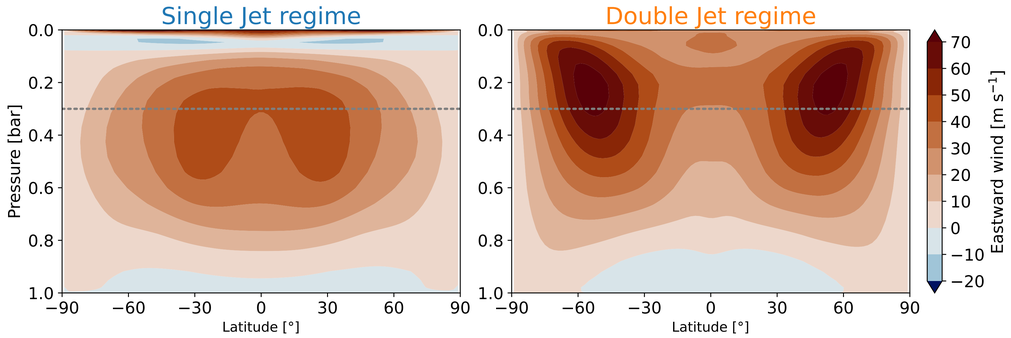
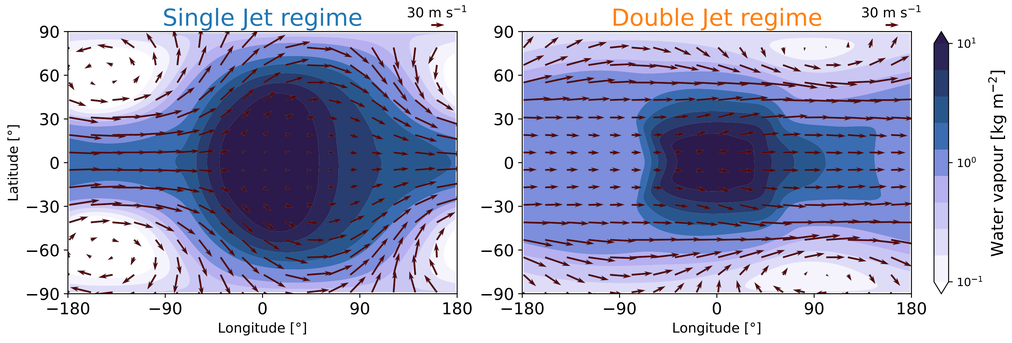
Background

- TRAPPIST-1e is one of the **best targets** for atmospheric characterisation and modelling.
- TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI) shows substantial differences between 3D climate simulations for moist atmospheres.
- Here we focus on the **emergence of two circulation regimes** and their **imprint in synthetic observations**.

3D simulations of atmospheric circulation

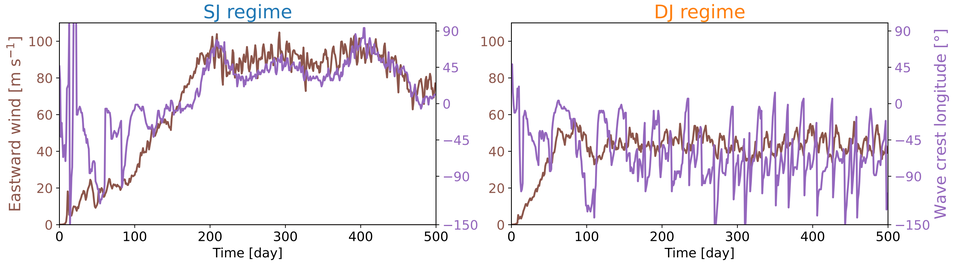
We use the **Met Office Unified Model (UM)** to simulate the atmospheric circulation, water cycle, and the resulting transmission spectrum. We use the orbital parameters of TRAPPIST-1e and assume it is **tidally locked** to its host star, an ultra-cool M-dwarf. We assume an atmosphere of **1 bar N_2**, 400 ppm CO_2, with H_2O as a condensible species. The surface is a slab ocean (**aquaplanet**). We run simulations with different initial conditions, physics schemes, and surface conditions. All of them develop one of the two regimes: **'Single Jet' (SJ)** or **'Double Jet' (DJ)**.



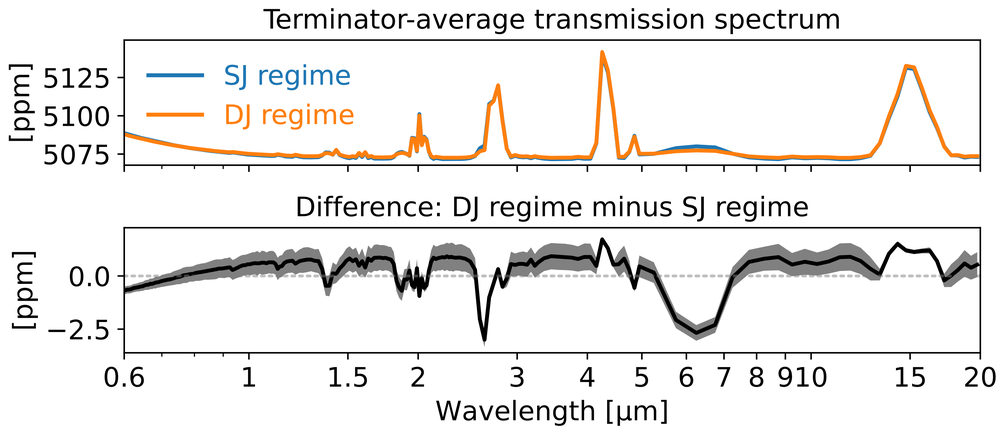


Emergence of superrotation

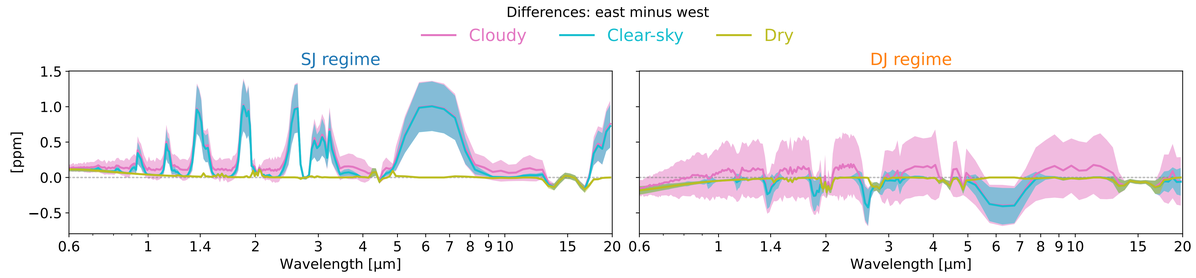
The **SJ regime** develops when the equatorial wind accelerates gradually and achieves resonance with the stationary Rossby wave, which in turn amplifies it. The **DJ regime** emerges when the initial wind acceleration is too rapid and the circulation splits into two mid-latitude jets, whilst the wave-jet resonance is not achieved.



Synthetic transmission spectrum



East-west terminator asymmetry



- The **SJ regime**: larger asymmetry than in the **DJ regime**, mostly due to water vapour.
- Continuum difference: ≈ 0.25 ppm, east terminator is cloudier in both regimes.
- Temperature contribution is the smallest (CO_2 bands and Rayleigh slope).

Summary

- **Two distinct** atmospheric circulation regimes emerge in the 3D simulations of TRAPPIST-1e.
- Model parameterizations and **even initial conditions** can trigger the regime transition.
- **Wave-jet resonance** during the model spin-up defines what regime emerges (**SJ regime**: yes, **DJ regime**: no).
- The two regimes have differences in the **transmission spectrum**, mostly due to **water vapour**, but they are too small to be observable with current technology.
- **Terminator asymmetry** is more pronounced in the **SJ regime** than that in the **DJ regime**.

More info

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